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b) Test for consistency and if possible, solve the following systems of equations by rank method :
$x - y + 2z = 2$, $2x + y + 4z = 7$, $4x - y + z = 4$

42. a) If $z = x + iy$ and $\arg\left(\frac{z-1}{z+2}\right) = \frac{\pi}{4}$, show that $x^2 + y^2 + 3x - 3y + 2 = 0$.

OR

b) Show that $\left(\frac{19+9i}{5-3i}\right)^{15} - \left(\frac{8+i}{1+2i}\right)^{25}$ is purely imaginary.

43. a) Solve the equations $6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$

OR

b) If $a_1, a_2, a_3, \ldots a_n$ is an arithmetic progression with common difference d, prove that $\tan\left[\tan^{-1}\left(\frac{d}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{d}{1+a_2a_3}\right) + \cdots + \tan^{-1}\left(\frac{d}{1+a_{n-1}a_n}\right)\right] = \frac{a_n - a_1}{1+a_1a_n}$

44. a) Show that the line $x - y + 4 = 0$ is a tangent to the ellipse $x^2 + 3y^2 = 12$. Also find the coordinates of the point of contact.

OR

b) For the ellipse $4x^2 + y^2 + 24x - 2y + 21 = 0$, find the centre, vertices, and the foci. Also prove that the length of latus rectum is 2.

45. a) Cross section of a Nuclear cooling tower is in the shape of a hyperbola with equation $\frac{x^2}{30^2} - \frac{y^2}{44^2} = 1$. The tower is 150m tall and the distance from the top of the tower to the centre of the hyperbola is half the distance from the base of the tower to the centre of the hyperbola. Find the diameter of the top and base of the tower.



OR

b) Prove by vector method that $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

46. a) If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}, \vec{b} = 3\hat{i} + 5\hat{j} + 2\hat{k}, \vec{c} = -\hat{i} - 2\hat{j} + 3\hat{k}$, verify that $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$

OR

b) Find the vector equation in parametric form and Cartesian equations of the line passing through $(-4, 2, -3)$ and is parallel to the line $\frac{-x-2}{4} = \frac{y+3}{-2} = \frac{2z-6}{3}$.

47. a) Find the image of the point (5,2,6) with respect to the plane $x + y - z = 9$

OR

b) Solve : $(x-4)(x-2)(x-1)(x+1) + 8 = 0$

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Class : 12

Register Number

COMMON QUARTERLY EXAMINATION - 2026-27

MATHEMATICS

Time Allowed : 3.00 Hours] [Max. Marks : 90

PART - I

1. Answer all the questions by choosing the correct answer from the given 4 alternatives 20 x 1 = 20
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. if $A\begin{bmatrix}1 & -2\\ 1 & 4\end{bmatrix} = \begin{bmatrix}6 & 0\\ 0 & 6\end{bmatrix}$, then A =

(a) $\begin{bmatrix}1 & 2\\ -1 & 4\end{bmatrix}$ (b) $\begin{bmatrix}1 & -2\\ 1 & 4\end{bmatrix}$ (c) $\begin{bmatrix}4 & -1\\ 2 & 1\end{bmatrix}$ (d) $\begin{bmatrix}4 & 2\\ -1 & 1\end{bmatrix}$

2. If $P = \begin{bmatrix}1 & x & 0\\ 1 & 3 & 0\\ 2 & 4 & -2\end{bmatrix}$ is the adjoint of 3 × 3 matrix A and $|A| = 4$, then x is

(a) 11 (b) 14 (c) 12 (d) 15

3. if $A = \begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$, and $A(\text{adj } A) = \begin{bmatrix}k & 0\\ 0 & k\end{bmatrix}$, then k =

(a) 0 (b) 1 (c) $\sin\theta$ (d) $\cos\theta$

4. The area of the triangle formed by the complex numbers z, iz, and z + iz in the Argand's diagram is

(a) $\frac{3}{2}|z|^2$ (b) $\frac{1}{2}|z|^2$ (c) $2|z|^2$ (d) $|z|^3$

5. If $\left|z - \frac{3}{z}\right| = 2$, then the least value of $|z|$ is

(a) 5 (b) 3 (c) 1 (d) 2

6. If z is a complex number such that $z \in \mathbb{C} \setminus \mathbb{R}$ and $z + \frac{1}{z} \in \mathbb{R}$, then $|z|$ is

(a) 2 (b) 1 (c) 3 (d) 0

7. If α, β, and γ are the zeros of $x^3 + px^2 + qx + r$, then $\sum\frac{1}{\alpha}$ is

(a) $-\frac{p}{r}$ (b) $-\frac{q}{r}$ (c) $-\frac{q}{p}$ (d) $\frac{q}{r}$

8. The number of real numbers in $[0, 2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1 = 0$ is

(a) ∞ (b) 1 (c) 2 (d) 4

9. The number of positive zeros of the polynomial $\sum_{r=0}^{n} {}^nC_r(-1)^r x^r$ is

(a) < n (b) r (c) 0 (d) n

10. The value of $\sin^{-1}(\cos x)$, $0 \le x \le \pi$ is

(a) $x - \frac{\pi}{2}$ (b) $\pi - x$ (c) $x - \pi$ (d) $\frac{\pi}{2} - x$

11. $\sin^{-1}(2\cos^2 x - 1) + \cos^{-1}(1 - 2\sin^2 x) =$

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

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12. The equation $\tan^{-1}x - \cot^{-1}x = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ has

(a) two solutions (b) no solution (c) unique solution (d) infinite number of solutions

13. The eccentricity of hyperbola whose latus rectum is 8 and conjugate axis is equal to half the distance between the foci is

(a) $\frac{4}{\sqrt{3}}$ (b) $\frac{4}{3}$ (c) $\frac{3}{2}$ (d) $\frac{2}{\sqrt{3}}$

14. The equation of the normal to the circle $x^2+y^2-2x-2y+1=0$ which is parallel to the line $2x+4y=3$ is

(a) $x+2y+3=0$ (b) $x+2y=3$ (c) $x-2y+3=0$ (d) $2x+4y+3=0$

15. Tangents are drawn to the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$ parallel to the straight line $2x-y=1$. One of the points of contact of tangents on the hyperbola is

(a) $\left(\frac{-9}{2\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ (b) $\left(\frac{9}{2\sqrt{2}},\frac{-1}{\sqrt{2}}\right)$ (c) $(3\sqrt{3},-2\sqrt{2})$ (d) $\left(\frac{9}{2\sqrt{2}},\frac{1}{\sqrt{2}}\right)$

16. Consider the vectors $\vec{a},\vec{b},\vec{c},\vec{d}$ such that $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})=0$. Let P_1 and P_2 be the planes determined by the pairs of vectors $\vec{a},\vec{b}$ and $\vec{c},\vec{d}$ respectively. Then the angle between P_1 and P_2 is

(a) 90° (b) 60° (c) 0° (d) 45°

17. If $\vec{a}=\hat{i}+\hat{j}+\hat{k}, \vec{b}=\hat{i}+\hat{j}, \vec{c}=\hat{i}$ and $(\vec{a}\times\vec{b})\times\vec{c}=\lambda\vec{a}+\mu\vec{b}$, then the value of $\lambda+\mu$ is

(a) 1 (b) 6 (c) 3 (d) 0

18. The distance between the planes $x+2y+3z+7=0$ and $2x+4y+6z+7=0$ is

(a) $\frac{\sqrt{7}}{2}$ (b) $\frac{7}{2\sqrt{2}}$ (c) $\frac{\sqrt{7}}{2\sqrt{2}}$ (d) $\frac{7}{2}$

19. For the ellipse below, the length of its latus rectum is $\frac{x^2}{25}+\frac{y^2}{9}=1$

(a) $\frac{18}{5}$ (b) $\frac{9}{5}$ (c) 8 (d) $\frac{16}{5}$

20. If z is $z=\frac{1+i}{1-i}$, then z^{2026} equals

(a) 1 (b) −1 (c) i (d) −i

PART - II

1. **Answer any 7 questions** — 7 x 2 = 14
2. **Each question carries 2 marks**
3. **Question number 30 is compulsory**

21. Solve the following system of linear equations by matrix inversion method: $2x-y=8,\ 3x+2y=-2$

22. Simplify : i^{1729}

23. If $z=x+iy$, find the following in rectangular form : $\mathrm{Re}(\bar{iz})$

24. Construct a cubic equation with roots : $2, \frac{1}{2}$ and 1.

25. Find the principal value of $\sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right)$

26. Find the domain of $\cos^{-1}\left(\frac{2+\sin x}{3}\right)$.

27. If $y=2\sqrt{2}x+c$ is a tangent to the circle $x^2+y^2=16$, find the values of c.

28. Find the equation of the parabola in each of the cases given below: vertex $(1,-2)$ and focus $(4,-2)$.

29. Find the volume of the parallelepiped whose coterminus edges are given by the vectors $2\hat{i}-3\hat{j}+4\hat{k}$, $\hat{i}+2\hat{j}-\hat{k}$ and $3\hat{i}-\hat{j}+2\hat{k}$

30. Find the intercepts cut off by the plane $\vec{r}\cdot(6\hat{i}+4\hat{j}-3\hat{k})=12$ on the coordinate axes.

PART - III

1. **Answer any 7 questions** — 7 x 3 = 21
2. **Each question carries 3 marks**
3. **Question number 40 is compulsory**

31. Find the inverse of the non-singular matrix $A=\begin{bmatrix}0 & 5\\ -1 & 6\end{bmatrix}$, by Gauss-Jordan method.

32. Find adj(adj (A)) if adj $A=\begin{bmatrix}1 & 0 & 1\\ 0 & 2 & 0\\ -1 & 0 & 1\end{bmatrix}$.

33. Obtain the Cartesian form of the locus of $z=x+iy$ in each of the following cases: $|z+i|=|z-1|$

34. Solve the equation : $x^4-14x^2+45=0$.

35. Find the value of $\tan^{-1}(-1)+\cos^{-1}\left(\frac{1}{2}\right)+\sin^{-1}\left(-\frac{1}{2}\right)$

36. Find the angle between line and the plane $\vec{r}=(2\hat{i}-\hat{j}+\hat{k})+t(\hat{i}+2\hat{j}-2\hat{k})$ and the plane $\vec{r}\cdot(6\hat{i}+3\hat{j}+2\hat{k})=8$.

37. Find the equations of tangent and normal to the ellipse $x^2+4y^2=32$ when $\theta=\frac{\pi}{4}$.

38. An engineer designs a satellite dish with a parabolic cross section. The dish is 5m wide at the opening, and the focus is placed 1.2 m from the vertex
   (a) Position a coordinate system with the origin at the vertex and the x -axis on the parabola's axis of symmetry and find an equation of the parabola.
   (b) Find the depth of the satellite dish at the vertex.

39. Show that the points (2,3,4), (−1,4,5) and (8,1,2) are collinear.

40. Show that $\cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right)=\sec^{-1}x, |x|>1$.

PART - IV

1. **Answer all the questions** — 7 x 5 = 35
2. **Each question carries 5 marks**

41. a) Solve the following system of linear equations, using matrix inversion method:
$2x+3y-z=9,\ x+y+z=9,\ 3x-y-z=1$ **(OR)**

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